IC101

Introductory Securities & Markets

Module Convenor: Professor George Alexandridis

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Your professors

Two of us teach this module, one in each half



Professor George Alexandridis

Professor of Corporate Finance, ICMA Centre

Research in mergers & acquisitions (M&A) and corporate finance.

Founder of start-ups and adviser to early-stage start-ups.

Senior adviser to an investment fund; consultancy across the industry.

Weeks 1 to 5 · Foundations of finance



Professor Brian Scott-Quinn

Emeritus Professor; founder of the ICMA Centre

Author of the module's textbook, and several others.

A career in investment banking and, today, fintech.

Adviser on the finance industry's transition to a sustainable economy.

Weeks 6 to 10 · Sustainable finance & ESG

What this module is, and how it works

What it's about

How the financial system actually works: markets, banks, money, risk and sustainability. The foundations under every finance job.

Textbook

Scott-Quinn, Commercial and Investment Banking (free via the Library).

Assessment

Written exam 50% (first half) + individual report 50%. Pass 40%.

The module runs in two halves

Weeks 1 to 5 · Prof. Alexandridis · Foundations

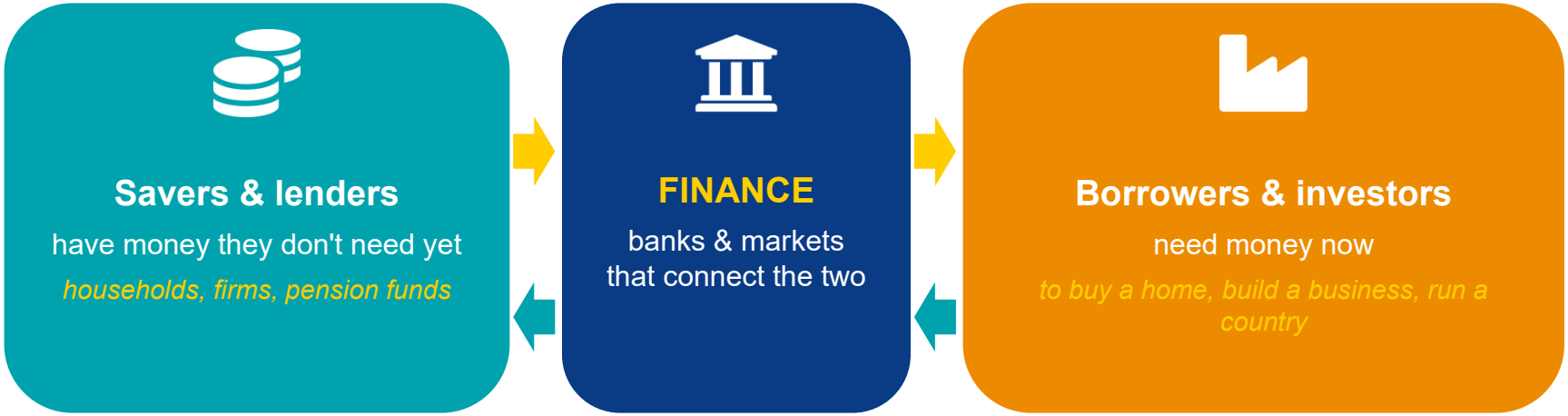
1. Pricing & the three pillars of finance
2. Balance sheets & financial intermediaries
3. Commercial, investment & central banks
4. Money & liquidity
5. Capital flows & the four sectors

Weeks 6 to 10 · Prof. Scott-Quinn · Sustainable finance

- ESG in banking & investment
- Climate-change risk & the energy transition
- **How finance can help, leading into your report**

What is finance, and what is it for?

At its simplest, it connects two sides and moves money between them

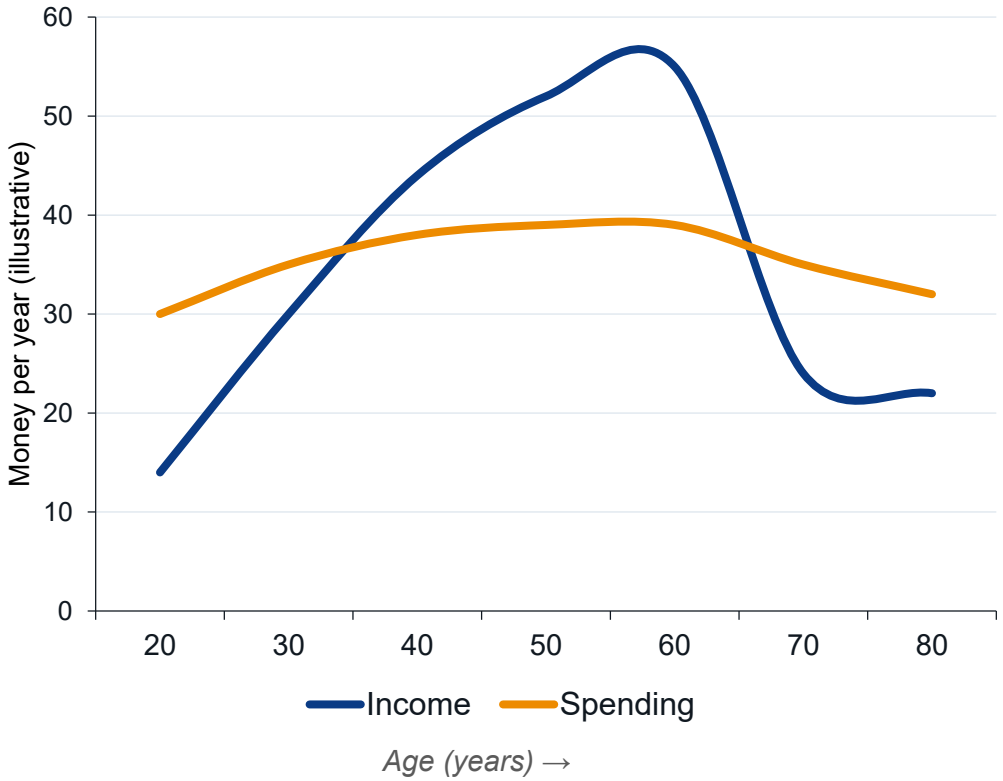


Capital flows one way (savers to finance to borrowers); **interest, profit and repayment flow back** the other way.

Its purpose: put the world's savings to work, send money to where it is most useful, and let everyone move money across time.

Why does finance exist? Start with you

One person's life: income humps up then drops at retirement, but you want steady spending



Three stages of life

Young: you spend more than you earn, so you BORROW.

Mid-life: you earn a bit more than you spend, so you can SAVE and INVEST.

Older: you live off your savings/pension (DISSAVE).

Finance lets you smooth spending across your life. That basic need is the root of the whole system.

Illustrative: one typical person across their own life, not average or real data. The classic 'life-cycle' picture, income humps up then drops at retirement while spending stays smooth.

The same need, scaled up

Companies and governments do exactly what you do



Companies

Raise money now to invest and grow: build factories, hire people, develop products. Then repay it from future profits.

Think of Tesla, Rolls-Royce and AstraZeneca

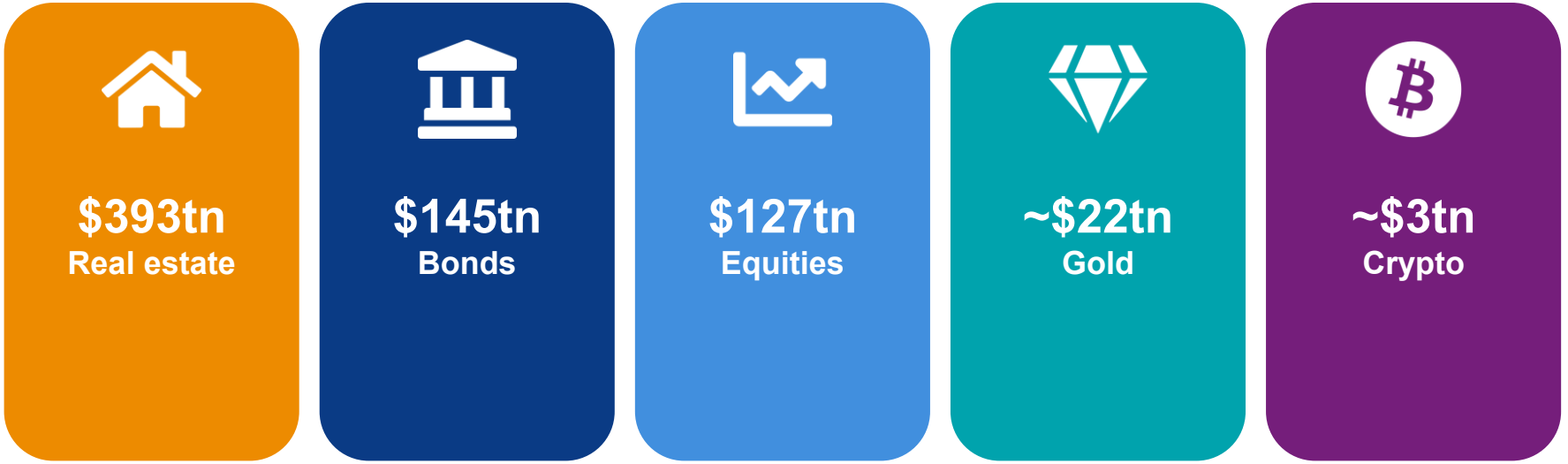


Governments

Borrow now (by issuing government bonds) to fund schools, hospitals/NHS and roads. Then repay from future taxes.

So finance channels the world's savings into productive investment. That is what creates growth, jobs and progress.

How big is finance? It's not just stocks & bonds



Real estate is the biggest store of wealth on earth, nearly 4× world GDP. Stocks and bonds are only part of the picture; add property, gold and crypto. **This whole universe needs pricing and managing.**

DEEPER READ All of the world's money and markets, in one picture (Visual Capitalist) ↗

Sources: Savills (real estate, 2024); SIFMA (bonds & equities, 2024); market estimates (gold, crypto).

The players. Map of the finance industry

Who has the money, who moves it, who needs it

Capital (£) flows this way →

← Securities & returns flow back the other way

Buy side

Asset owners: Savers & investors, **HAVE the money**.

e.g. Asset, pension & hedge funds

Norges Bank Investment Management · BlackRock · Schroders · BRIDGEWATER · Vanguard · abrdn · CITADEL · lgps Local Government Pension Scheme · L&G · Man

Sell side

Banks & markets, **MOVE the money**.

e.g. Commercial & investment banks

Goldman Sachs · BARCLAYS · J.P.Morgan · HSBC · LAZARD

Issuers

Companies & governments, **NEED the money.**

e.g. corporates, governments

bp · HUAWEI · HM Treasury

Around this core: **accountants & consultants** advise, **central banks, regulators & exchanges** keep it running, and **fintech** reshapes it.

Where this can take you

ICMA Centre graduates, and what the roles are really worth



50-60% of ICMA Centre graduates go straight into finance, the rest into consulting, accounting, real estate and tech. London grad analysts ~£32k-£70k.

AI is reshaping finance, and your first job

What is changing, and what to focus on

What it means for you

AI now does much of the entry-level number-crunching that used to be a graduate's first job.

So there are fewer purely analytical junior roles, and far more value on judgment, client trust, and asking the right questions.

Focus now on: analytical thinking, AI fluency, and communication.

Where the work is heading

AI takes over

- gathering data
- first-draft models
- routine reporting
- summarising documents

You must own

- communication
- judgment
- client trust & ethics
- asking the right question
- spotting when AI is wrong

Evidence: the WEF Future of Jobs 2025 finds 39% of workers' core skills will change by 2030; the fastest-growing skills are AI & big data, and **analytical thinking** is the #1 skill employers want.

DEEPER READ WEF Future of Jobs 2025: the skills you will need, with the evidence ↗

So learn to prompt, and to judge

Writing a precise prompt and checking the answer is now a core skill

Why it matters

Anyone can type a lazy question. The graduates who stand out write precise, evidence-based prompts, then judge and improve the answer.

Every 'Explore with AI' button in these slides is a worked example. Run it, then make it your own.

Anatomy of a good prompt

ROLE	Act as a patient finance tutor.
CONTEXT	I'm a first-year finance student, new to bond pricing.
TASK	Explain, with a £100 example, why bond prices fall when rates rise.
EVIDENCE	Base it on real data and cite a reputable source.
FORMAT	Step by step, then quiz me with two questions.

CLICK **Try this exact prompt, then make it your own** ↗

A lazy prompt gets a lazy answer.

Quick poll, grab your phones

Would you rather have…

A £1,000 right now

B £1,100 in one year's time

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The three pillars of finance

Price = future cash flows, discounted for TIME (what it's worth now) and RISK (a premium for uncertainty).



1. Time value of money

A pound today is worth more than a pound tomorrow.

Your poll: £1,000 now or £1,100 next year? Interest and inflation.



2. Risk management

Returns are never certain, risk must be measured & managed.

What if that £1,100 might not arrive? The extra you'd demand is the risk premium.



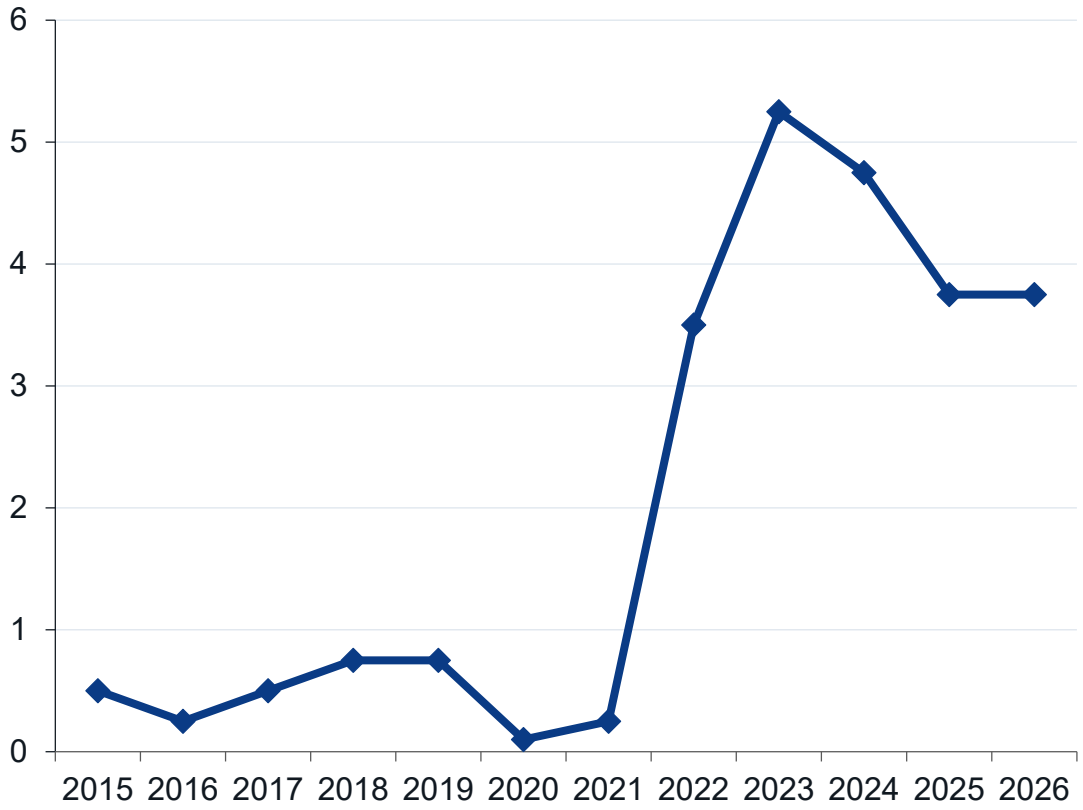
3. Asset pricing

Combine the first two to price bonds, shares, property…

Learn these once, they price everything, for the rest of your career.

First, the price of money itself: interest rates

Set by the Bank of England, this is what it costs to borrow, and it moves everything



In 18 months

money went from almost free (0.1%) to the priciest in 15 years (5.25%).

When this one line moves, so does the whole economy:

- Borrowing: loans, mortgages, cards dearer
- Saving: cash finally earns a return
- Bonds: prices fall as rates rise
- Houses & shares: cool as credit bites
- The pound: higher rates tend to lift it
- The goal: cooler demand, lower inflation

And a higher rate makes money today worth even more, that is Pillar 1.

DEEPER READ How a change in interest rates ripples through the whole economy (ECB explainer)

Source: Bank of England, Bank Rate 2015-2026 (year-end).

Pillar 1 · The time value of money

Let's put real numbers on the poll

£1,000 now, or £1,100 in a year?

If you could earn 5% on your money:

- £1,000 today grows to **£1,050** in a year.
- So £1,100 next year is worth:

£1,100 ÷ 1.05 = £1,047 today.

B wins, but if we change the rate the answer may be different.

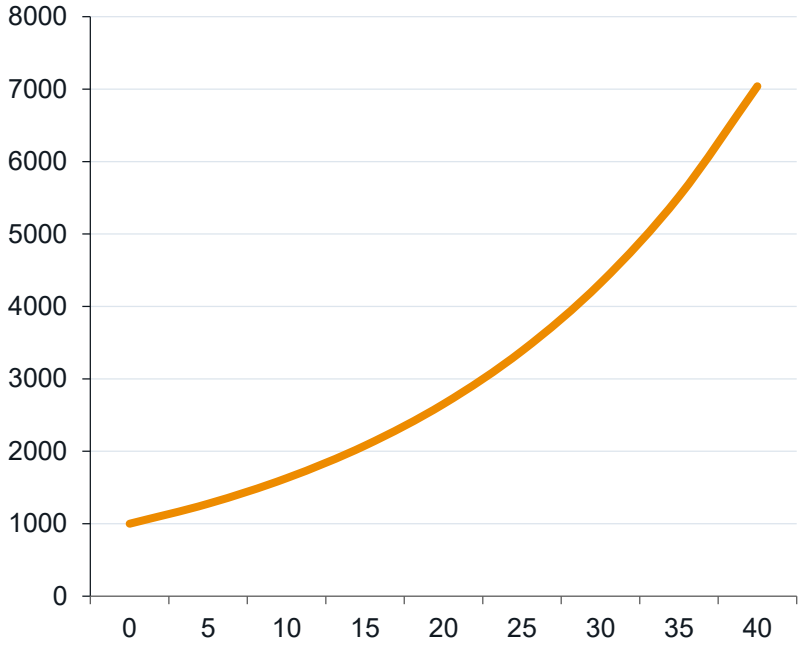
PV = future cash ÷ $(1 + r)^t$

PV: Present value

r: discount rate (opportunity cost)

t: number of periods (in this case 1)

That's discounting and it's how we value everything.



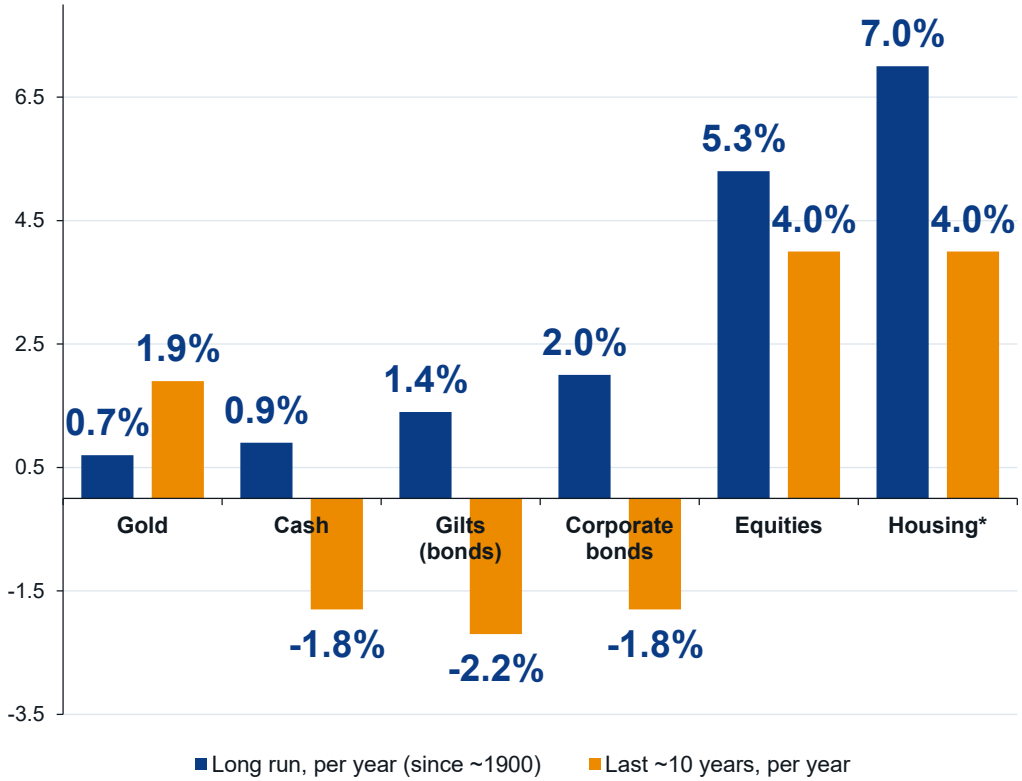
The flip side, compounding: £1,000 at 5% becomes £7,040 over 40 years.

$FV = PV \times (1 + r)^t = 1{,}000 \times 1.05^{40}$

CLICK Explore the time value of money with AI

Why money has to be put to work

Real (after-inflation) returns per year: the long run vs the last decade



Safe is not the same as free

In the last decade, cash, gilts and corporate bonds all LOST money after inflation.

Over a lifetime, where you put your money is the biggest financial decision you'll make.

Higher return, though, comes with more risk. That trade-off is Pillar 2.

*Sources: Barclays Equity Gilt Study (cash, gilts, equities); iShares UK corporate-bond fund; gold LBMA (GBP); ONS/Nationwide housing. Long run since ~1900 (housing 1870-2015); recent ~10 years to 2022. *Housing = total return incl. rent, indicative.*

Quick poll, grab your phones

You have £1,000 to invest for 10 years. Where does it go?

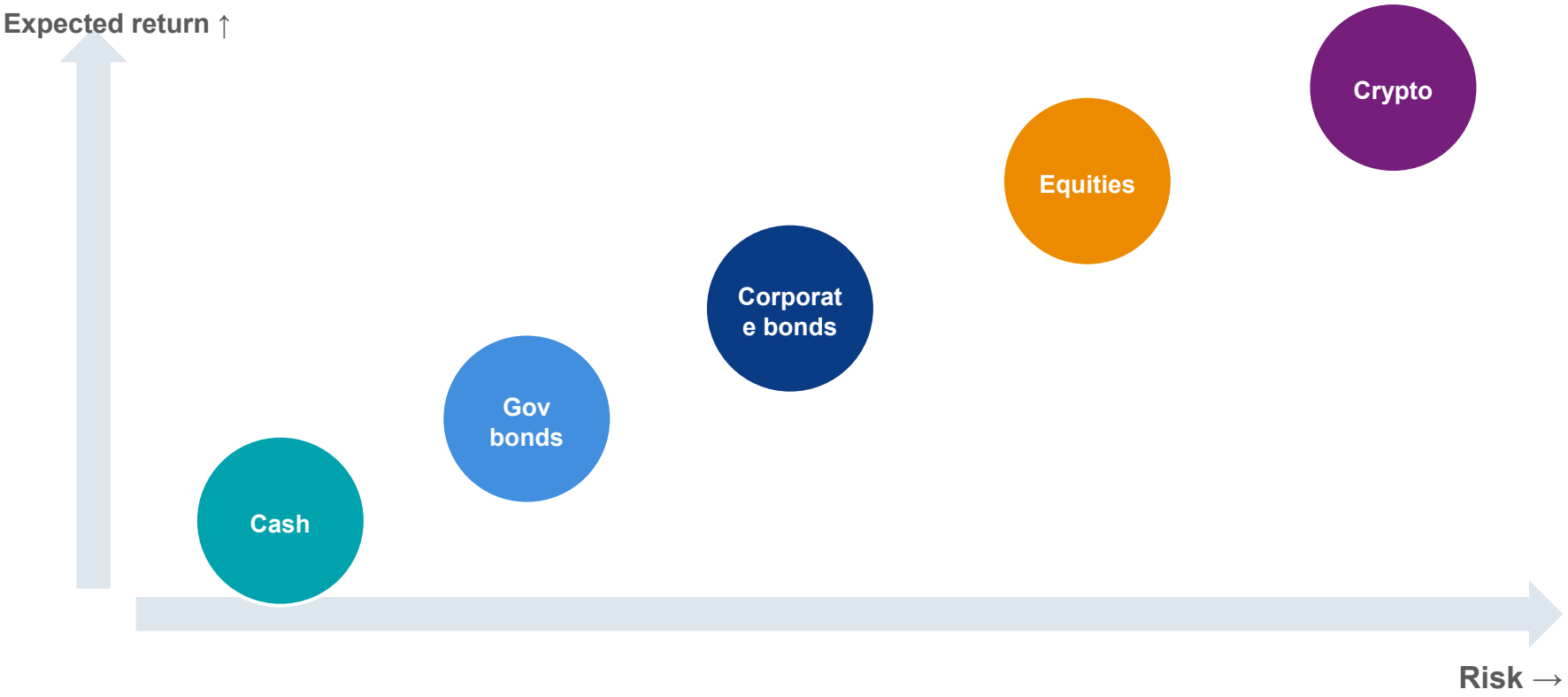
- **A** Leave it in a savings account
- **B** Government & corporate bonds
- **C** A global shares (equity) fund
- **D** Crypto
- **E** A mix of the above

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Pillar 2 · No return without risk

Investors are risk-averse, they take more risk only if they expect more return.

And the ACTUAL return you get is rarely the return you EXPECTED. That gap is risk.



Three ways to tame risk

Diversification



One basket:
one shock wipes you out

Many baskets:
one shock barely hurts

Don't put all your eggs in one basket.
Spread money across many assets.

Hedging

Take an offsetting position, so a loss on one side is cancelled by a gain on the other.

(e.g derivatives/hedge funds)

Insurance

Pay a known premium to hand a risk you can't afford to someone else (e.g. CDS during the 2008 financial crisis)



Pillar 3 · What are we actually pricing?



Capital assets

Produce cash flows.

e.g. Bonds, shares, loans

Priced by discounting its cash flows.

Derivatives

Value derived from something else.

e.g. Options, futures, swaps

Priced off the underlying asset it tracks.

Real assets

Physical, with intrinsic use.

e.g. Property, commodities

Priced by comparable sales and rental yield.

Speculative

Worth only what the next buyer pays.

e.g. Some crypto, art

Only the next buyer sets the price.

Derivatives in real life: a fuel-price futures contract lets an airline lock in its jet-fuel cost; an Apple put option insures your shares against a fall; an interest-rate swap turns a floating loan into a fixed one.

Whatever the asset, the price always comes back to **cash flows, risk and time.**

Discounted cash flow: the formula

Your first time seeing it, so let's build it up piece by piece

$$PV = CF_1 / (1+r)^1 + CF_2 / (1+r)^2 + \ldots + CF_n / (1+r)^n$$

$$= \Sigma \; CF_t / (1+r)^t \qquad (\Sigma \text{ just means: add up every year, } t = 1 \text{ to } n)$$

PV
value today of future cash

CF
cash received each period

r
the discount rate you demand (time + risk)

n
the period: year 1, 2, … n

In words: add up every future cash flow, but shrink each one, because money later is worth less than money now.

Next slide puts this to work: add up three years of an Apple project's cash, each shrunk by its own (1+r) factor.

CLICK **Explore the DCF formula with AI** ↗

Asset pricing in action (1): discounted cash flow

The same method values a single project or an entire company

One tool, two uses

Any asset is worth the future cash it produces, each pound pulled back to today.

A project: add up its few years of cash.

A whole company: the very same sum, just many more years of cash flows.

The 8% is the discount rate: the return Apple could earn elsewhere on equally risky money, its opportunity cost. *we take one Apple project, to keep the numbers small and clear.*

The calculation, step by step

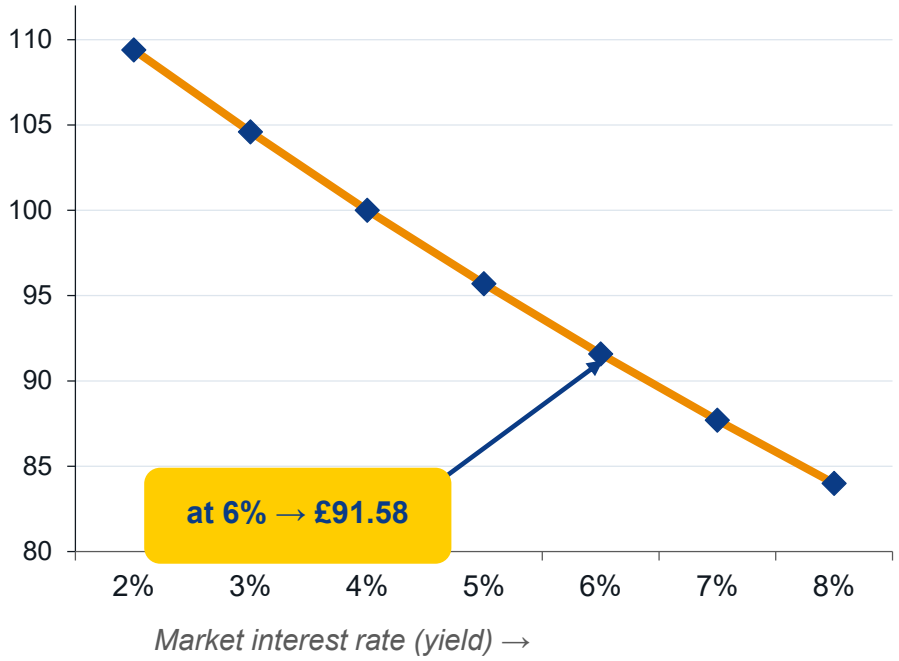
Apple project · $4bn cash in each year · costs $10bn today · discount rate 8%			
Year 1	$4.0bn	÷ 1.080	**$3.70bn**
Year 2	$4.0bn	÷ 1.166	**$3.43bn**
Year 3	$4.0bn	÷ 1.260	**$3.18bn**
Total value of that cash, today			**$10.31bn**
Less: cost to build it today			**- $10.00bn**
Net value created (NPV)			**+ $0.31bn**

Where the factors come from: year 3 = $4bn ÷ $(1.08)^3$ = $4bn ÷ 1.260 = $3.18bn.

Why 12% and not 8%? A riskier project, or higher interest rates, means investors demand more. Same cash flows, shrunk harder: total falls to $9.61bn, NPV becomes -$0.39bn. Don't invest. The rate decides.

Asset pricing in action (2): a bond

Same idea: a bond's price is its future coupons plus face value, discounted



When rates rise, a fixed £4 coupon looks worse than new bonds, so the old bond is worth less.

The bond price formula

Price = Σ coupon/(1+y)^{t} + face/(1+y)^{N}

Discount every £4 coupon and the £100 face at the market yield y.

Worked example at y = 6%

Coupons: £4/1.06 + … + £4/1.06^{5} = £16.85

Face: £100/1.06^{5} = £74.73

Price = £16.85 + £74.73 = £91.58

Exam-style question

Rates rise from 4% to 6%. This bond's price…

A) rises B) falls C) unchanged

Answer: B. Price falls from £100 to £91.58.

CLICK Explore bonds and interest rates with AI

Quick poll, grab your phones

Which is riskier to own?

A A US corporate bond (a big US company)

B A Malaysian government bond

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Asset pricing in action (3): shares

How much do you pay for £1 of a company's profit?

The P/E ratio

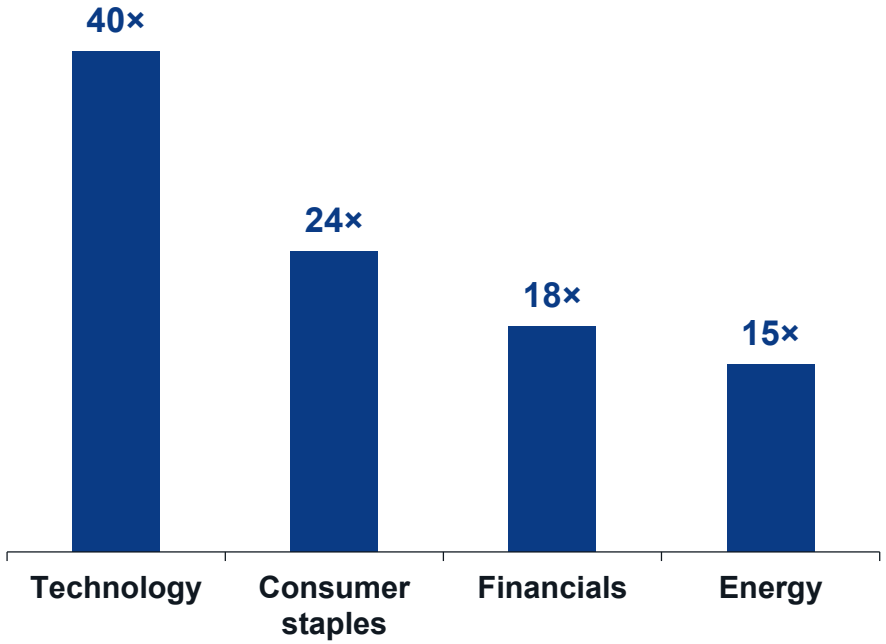
Price-to-earnings = share price ÷ earnings per share (EPS).

Worked example, Apple (2026):

$333 share price ÷ $8.29 EPS ≈ P/E of 40.

You pay ~$40 for every $1 of annual profit.

A high P/E = the market expects fast growth. Also used: price-to-book (price vs net assets).



The market pays ~40× for tech profits but ~15× for energy, it's paying for growth.

DEEPER READ What high valuations tell us about future returns: the Shiller CAPE ratio

Sources: company filings (Apple, 2026); Siblis Research sector P/E (2025).

So… can you trust the price?

The Efficient Market Hypothesis

The Efficient Market Hypothesis (EMH)

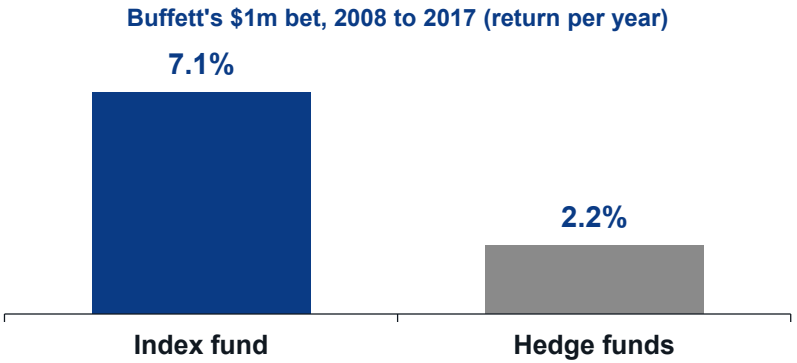
Prices already reflect all known information, so it's very hard to consistently beat the market.

Arbitrage, the law of one price, enforces it: if the same asset is cheap in one place, traders buy it until the gap closes.

But 'efficient' is not 'always right'. Fear and euphoria still pull PRICE away from VALUE.

"Markets can stay irrational longer than you can stay solvent."

John Maynard Keynes



DEEPER READ Warren Buffett's $1m bet: simple index funds beat the hedge funds, 7.1% vs 2.2% a year

The highlights, know these cold

1. Finance moves resources across time; price is the signal that allocates them.
2. Price vs value, and the three pillars: time value, risk management, asset pricing.
3. Time value: PV = future cash ÷ $(1+r)^t$, £1,100 next year is only ~£1,047 today at 5%.
4. Risk & return: investors are risk-averse; tame risk by diversification, hedging, insurance.
5. Pricing assets: DCF for companies; rates up → bond prices down; shares on P/E & price-to-book.
6. Asset types: capital, derivative, real, speculative.
7. Markets are broadly efficient (EMH) & arbitrage enforces one price, but not always rational.

Your written exam is built from this slide.

Exam check, can you answer these?

Same style as your exam

If market interest rates RISE, the price of an existing fixed-coupon bond will…

A) rise B) fall C) not change

Answer: B. A fixed coupon is worth less when new bonds pay more.

£100 received in one year is worth, today…

A) more than £100 B) less than £100 C) exactly £100

Answer: B. You could invest £100 today and have more than £100 next year.

Company X trades at a P/E of 40, company Y at 15. The market most likely…

A) expects X to grow faster B) thinks X is cheaper C) is wrong

Answer: A. A higher P/E means investors pay more per £1 of profit, expecting growth.

Where we are, and where we're going

Lecture 1 of 5

1	**The big picture & the three pillars**	✓ today
2	**Balance sheets & financial intermediaries**	→ next week
3	Commercial, investment & central banks	
4	Money & liquidity	
5	Capital flows & the four sectors	

Today you built the toolkit for valuing anything. **Next: the balance sheets and banks that actually use it to move money through the economy.**

Over to you

Before next week

- **Read Chapter 1 of the textbook.**
- Look ahead: Chapters 2 & 3.
- Take the Blackboard quiz.
- Explore ICMAC UG Programmes Blackboard page (School information)
- Find and read two relevant articles in the FT this week.
- Log-in to LSEG Workspace in one of the dealing rooms (you will have to use it for your project)



Keep thinking

We polled the Malaysian vs US bond question, now find the actual credit ratings of each.

Were you right?

YOUR SEMINAR THIS WEEK **Seminar 1: Risk & Return**

Go, it goes deeper on today's lecture: the price of credit and why rising rates push bond prices down, the life-cycle, and the three pillars with worked NPV examples.

To prepare: read Chapter 1 and come with questions.